

SHAYAN BHOWMIK

Symbiosis Institute of Technology — B.Tech. CSE — 2024 – 2028 — GPA: 8.17/10

☎ +91-9835116087 ✉ shayan.bhowmik05@gmail.com [in LinkedIn](#) [GitHub](#) [Portfolio](#)

Introduction

Design, build, and ship production-grade AI systems and full-stack applications, spanning system architecture, RAG pipelines, and scalable backends. CSE undergraduate with strong fundamentals in applied ML and open-source engineering, building real-world systems through research and hackathons.

Skills

Languages: Python, Java, C++, TypeScript

Frontend: React, Vite, Tailwind CSS, HTML, CSS

Backend: Node.js, Express.js, FastAPI, Flask

Databases: PostgreSQL, MySQL, Supabase

Tools & DevOps: Git, GitHub, Vercel, PyTorch, Postman, Linux, Docker

Soft Skills: Leadership, Problem-Solving, Ownership, Communication

Experience

Kytes

June 2026 – July 2026

Software Engineer Intern

Internship

- Architected an autonomous risk-response agent using LangGraph and OpenRouter, applying PMI/PMBOK strategies (Avoid, Transfer, Mitigate, Accept) to automate project risk decision-making.
- Engineered prompt frameworks and LLM orchestration logic for context-aware risk analysis, integrating a SQLite-backed pipeline for lightweight, low-latency data persistence.
- Collaborated with the engineering team to deploy the agent end-to-end, from backend architecture to production deployment on Streamlit Cloud.

Google

May 2026 – August 2026

Student Ambassador

Internship

- Represented Google Gemini on campus, driving developer-tool adoption among students through workshops, events, and technical content.

Projects

RAG Pipeline with Hybrid Search | *FastAPI, Next.js, Supabase*

2026 – Present

- Built a document ingestion pipeline for a hybrid RAG system, implementing semantic chunking, sentence-transformer embeddings, and BM25 sparse indexing over a document corpus.
- Designed the Supabase/pgvector schema and a unified ingestion endpoint to populate dense and sparse indexes end-to-end from raw documents in a single call.
- Currently extending the system with Reciprocal Rank Fusion (RRF), cross-encoder reranking, and Claude API-based grounded generation.

Deep Learning Model for Brain Tumor Detection | *Python, PyTorch*

Jan 2026 – Apr 2026

- Developed a custom CNN for binary brain tumor classification using a 2.5D representation, stacking consecutive axial slices from 786 patients across BRATS 2020 and OASIS to preserve inter-slice spatial context.
- Designed a patient-level evaluation pipeline with leakage-free splits, class-weighted loss, and probability aggregation with calibrated threshold selection, achieving >99% slice-level accuracy and 100% patient-level accuracy on the held-out test set (ROC-AUC 0.999).
- Applied Grad-CAM with brain-region masking for model interpretability and conducted ablation studies quantifying the contributions of 2.5D context, normalization, and class-weighted loss.

Awards & Certifications

- **Claude Code 101** — *Anthropic* Jul 2026
- **Introduction to Generative AI** — *Google* Dec 2025
- **Introduction to Prompt Engineering** — *Microsoft* Dec 2025